LECTURER: JOHN DOE

REINFORCEMENT LEARNING

REINFORCEMENT LEARNING
TOPIC OUTLINE

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Dynamic Programming and Markov Decision Processes	2
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UNIT 3

BANDIT PROBLEMS

STUDY GOALS



- Define **bandit problem**'s role in sequential decisions
- **Compare** ε-greedy, softmax, and index **policies**
- Apply **ε-greedy strategy** to solve bandit problems
- Evaluate optimal policy selection in **index-based bandits**

EXPLAIN SIMPLY



1. Explain how bandit problems relate to making optimal sequential decisions in various scenarios

2. Describe the key differences between ε-greedy, softmax, and index policies for solving bandit problems

3. Discuss the factors that influence the choice of an appropriate bandit policy in different applications

Dilemma: balancing immediate gains vs. exploration for long-term optimal choices in sequential decision-making

- **Scenario:** conduct clinical trials for multiple treatments



- **Problem:** how to balance optimal individual patient care and discover the overall best medical treatment?





How much to explore / exploit?

Source of the text: Freedman, 1987

Goal: balance optimal decisions with exploration in sequential processes

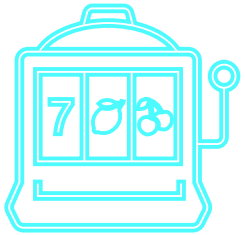


- **Adaptation**: optimize decisions based on evolving data
- **Iterative Learning**: continuously refine strategies for maximized gains

Applications

- Healthcare
- Finance
- Online advertising
- Recommender systems

ILLUSTRATION OF THE ONE ARMED BANDIT PROBLEM

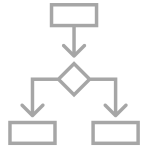


How to select the arm for optimal rewards?

- **Choice**: Selecting arms (actions) from the slot machine



- **Decision**: Repeatedly pulling arms to maximize cumulative rewards



- **Goal**: Achieving highest total reward over multiple pulls



- **Policy**: Strategy determining which arms to pull for optimal rewards



THE ONE ARMED BANDIT PROBLEM

- **Stages**: sequential units in the decision process
- **Action:** choice from sources of reward
- **Decision and rewards:** made and accrued within stages
- **Discount factor:** weights successive rewards

Data from the process history

$$n_{k,j} = \sum_{i=1}^{j} I\{a_j = k\}$$

Number of times action k is chosen

Expected total reward

$$R = \sum_{j=1}^{\infty} \sigma_j E[r_j]$$

Source of the text: Sutton and Barto, 2018.

THE ONE ARMED BANDIT PROBLEM

- **Optimal policy**: achieves maximum total reward

Optimal Total Reward

$$R^* = \sum_{j=1}^{\infty} \sigma_j \, \mu^*$$

- **Regret:** gap between optimal reward and chosen action's cumulative reward

$$\Delta^{\Phi} = R^* - R^{\Phi}$$

Realized Reward

$$R^{\Phi} = \sum_{j=1}^{\infty} \sigma_j \, E\,[\mu_{a_j}]$$

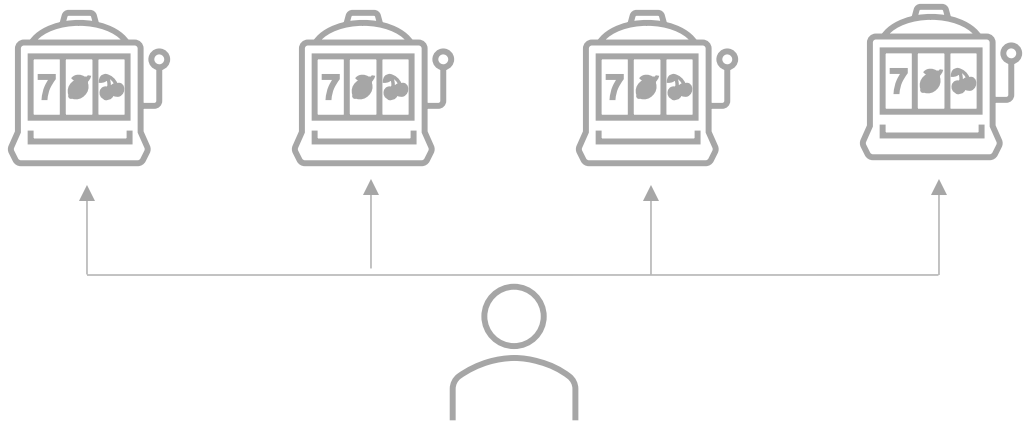


How can we **maximize gains** and **minimize regret?**

Source of the text: Sutton and Barto, 2018.

Multiple Arm Bandit: optimize choices among multiple actions to maximize cumulative rewards

- **Multiple Arms:** set of actions
- **Reward Distribution:** uncertain rewards for each action
- **Initial Knowledge**: limited or no reward distribution information
- **Learning**: update action preferences based on observed rewards over time



Maximize rewards:
How to choose actions?

THE EXPLOITATION VS EXPLORATION TRADE OFF

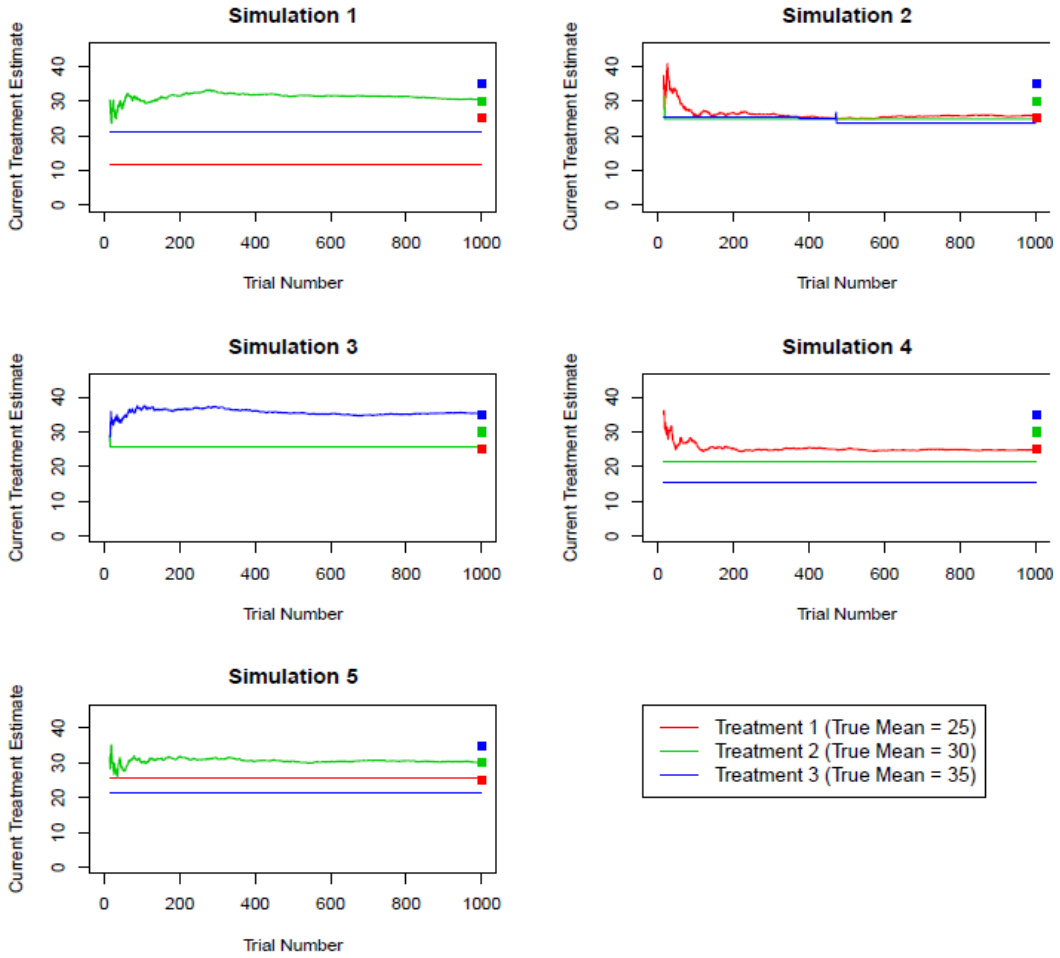
Exploitation (focus on known)

- Select action with highest current reward estimate
- Maximizing gains from actions with established rewards



Exploration (search for more)

- Try less-known actions to gather more information
- Balancing immediate gains with uncovering potentially better options



Computer simulation of the bandit problem

Source of the image: Own representation

EXPLOITATION - EXPLORATION ALGORITHMS

ε – Greedy

- Explores with probability ε, exploits otherwise
- Simple to implement and interpret
- Can get stuck in suboptimal actions

Softmax Action Selection

- Chooses actions probabilistically based on their estimated values
- Temperature parameter controls exploration level
- Gradually shifts focus from exploration to exploitation

Gradient Policy

- Probabilistically selects actions based on preferences
- Updates preferences using reward feedback
- Balances exploration with expected reward optimization

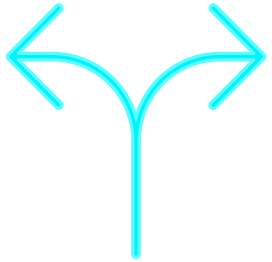


Trade-off: balancing exploration and exploiting known rewards

Source of the text: Sutton and Barto, 2018.

Action Value Methods

- Estimate rewards, explore for improved rankings
- Optimal performance by selecting the best arm
- Example: ε – greedy, softmax action selection, gradient policy



Index Methods

- Assign index values to actions based on exploration-exploitation balance
- Choose action with highest index value

$$Index = [Greedy] + [Uncertainty]$$

Confidence intervals

- Estimate arm properties from data
- Range of values with confidence

Confidence bounds

- Set boundaries for arm performance
- Define limits with confidence

UCB1 Index Method

- Initialize all arms as unexplored
- Calculate upper confidence bound
- Select arm with highest value
- Update arm statistics and repeat



Index methods guide action selection through value-driven exploration

REVIEW STUDY GOALS



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SESSION 3

TRANSFER TASK



Case study



An online news platform seeks to maximize user engagement by recommending the most relevant articles while balancing click-through rates and diversity of content.
The task involves selecting articles to display to users in order to optimize engagement and satisfaction, considering variables like user preferences, article popularity, and novelty

Task

Research how the multi-arm bandit problem can enhance article recommendation. Define arms, rewards, exploration, exploitation

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your results.

The results will be discussed in plenary.





1. five-stage finite horizon four-arm bandit problem uses the following sequence of discount factors: $\rho[k] = 1, 1, 1, 0.5, 0.25$, $k = 1,2,3,4,5$. Suppose the true mean rewards are 24.5, 36.5, 19.0, 32.5, for arms $j = 1,2,3,4$ respectively. What is the maximum possible total reward?
   a) 71.25
   b) 95.0
   c) 136.875
   d) 182.5



2. A three-arm bandit problem is a 100-stage finite horizon process. The true mean rewards of the arms are 10.1, 9.0 and 7.5. Suppose a policy selects each arm for ten stages, calculates the average reward for each, then selects the arm with the highest observed average. If the optimal reward is correctly identified, what will be the regret?
   a) 37.0
   b) 266.0
   c) 973.0
   d) 1010.0



3. For softmax action selection, what best describes the policy as the temperature parameter approaches zero?
   a) An exploratory policy
   b) A greedy policy
   c) An ϵ-greedy policy
   d) A UCB1 policy

LIST OF SOURCES

Freedman, B. (1987). Equipoise and the ethics of clinical research. New England Journal of Medicine, 317 (3), 141–145.

Sutton, R. S., & Barto, A. G. (2018a). Reinforcement learning: An introduction (2nd ed.). MIT press.

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